STACKS, DEFORMATION THEORY: LEMMA-LIFT-MAP-COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5714-5731 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5733-5750, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 45 of queue_defos.json (pb=613, stmt 5714-5731, proof 5733-5750).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0642: sources/stacks-defos/pinned-original-source.tex, source lines 5750–5750. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings. Assume given the following data*

(1) *a complex of $\mathcal{O}$-modules $\mathcal{F}^\bullet$,*
(2) *a complex $\mathcal{K}_0^\bullet$ of $\mathcal{O}_0$-modules,*
(3) *a quasi-isomorphism $\mathcal{K}_0^\bullet \to \mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{O}_0$,*

Then there exist a quasi-isomorphism $\mathcal{G}^\bullet \to \mathcal{F}^\bullet$ such that the map of complexes $\mathcal{G}^\bullet \otimes_{\mathcal{O}} \mathcal{O}_0 \to \mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{O}_0$ factors through $\mathcal{K}_0^\bullet$ in the homotopy category of complexes of $\mathcal{O}_0$-modules.

Proof. Set $\mathcal{F}_0^\bullet = \mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{O}_0$. By Derived Categories, Lemma 0642 there exists a factorization

$$\mathcal{K}_0^\bullet \to \mathcal{L}_0^\bullet \to \mathcal{F}_0^\bullet$$

of the given map such that the first arrow has an inverse up to homotopy and the second arrow is termwise split surjective. Hence we may assume that $\mathcal{K}_0^\bullet \to \mathcal{F}_0^\bullet$ is termwise surjective. In that case we take

$$\mathcal{G}^n = \mathcal{F}^n \times_{\mathcal{F}_0^n} \mathcal{K}_0^n$$

and everything is clear. □

Date: stacks-project:defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.